

Algorithms for Deciding the Safety of States in Fully Observable Non-deterministic Problems

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Foundations of AI
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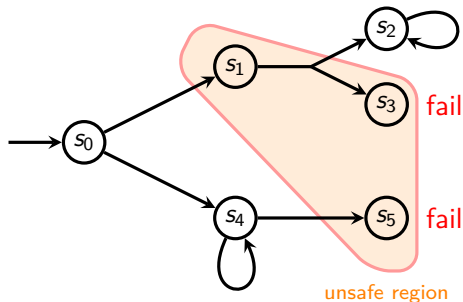
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Setting

We consider Fully Observable Non-Deterministic (FOND) planning but with **fail states** instead of goal states — solutions are policies $\pi : S \rightarrow A$ that avoid fail states

A state is **unsafe** if reaching fail state is unavoidable.



Given a state, we want to decide if it is **unsafe** or not.

Some Context

- We want to decide if states are **unsafe** to find faults in learned policies [Cascioli et al., 2025].
- Our problems are encoded in JANI [Budde et al., 2017].

Current Status

Jain et al. [2025] tested VI, iLAO*, LRTDP, and **TarjanSafe**.

👉 TarjanSafe worked very well.

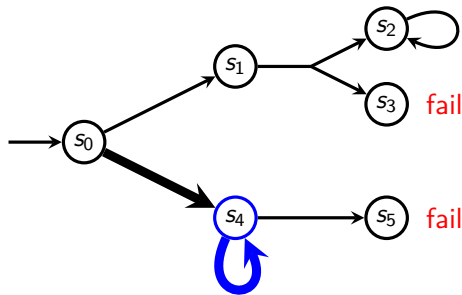
Our contributions

- TarjanSafe has an exponential worst-case behaviour w.r.t. graph
- we give a polynomial alternative

TarjanSafe [Jain et al., 2025]

DFS with SCC detection á la Tarjan's Algorithm

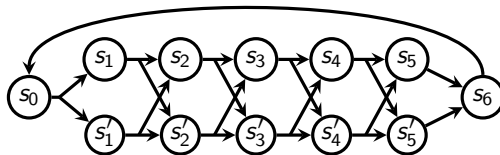
Marks SCC safe when popping SCC root and none if its elements are unsafe



TarjanSafe [Jain et al., 2025]

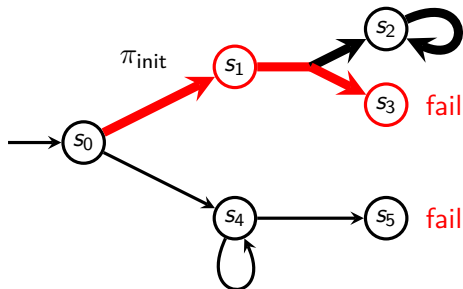
- best case $\Theta(|\pi_{\text{min-safe}}|)$
- worst case $\Omega(2^m)$

$m = |A|$, $|\pi_{\text{min-safe}}|$ is a
min. safe policy size
(if safe π exists)



Naïve and Improved Policy Iteration (nPI and iPI)

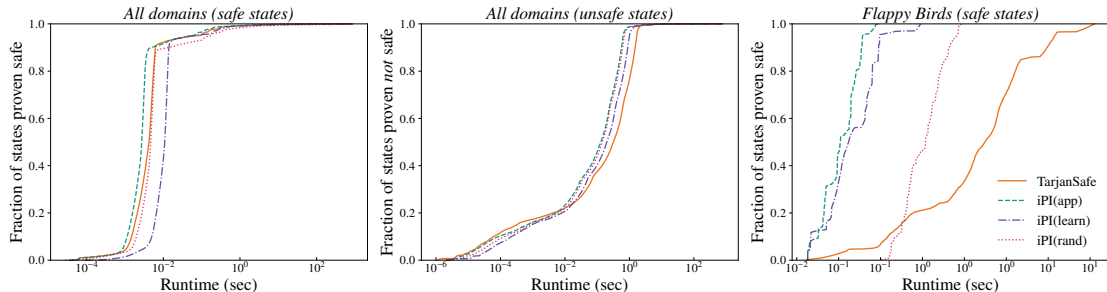
Start with π_{init} , check it with DFS, propagate issues and update π to fix them, repeat.



- best case $\Theta(|\pi_{\text{min-safe}}|)$
- worst case $O(bmn)$

$m = |A|$, $n = |S|$,
 $b = \text{non-det. branching}$,
 $|\pi_{\text{min-safe}}|$ is a min. safe
policy size

- **Domains:** blocksworld, transport, control benchmarks (e.g., inverted pendulum), *Flappy Bird*
- **Algorithms:** TarjanSafe and iPI with different π_{init}



- Over “All domains”: similar performance (these are TarjanSafe-friendly domains)
- Over “Flappy Bird”: exponential gap (for deciding safe states)
- Different π_{init} : difference is probably implementation

Summary

- **Setting:** decide safety of states in FOND
- TarjanSafe fast in practice but exponential worst-case runtime
- our algorithm (iPI) is fast in practice and polynomial worst-case!

In the paper we also look at **Unsafety Propagation**, connection between Goal-Reaching and State-Avoiding FOND, and a bit more

For more:

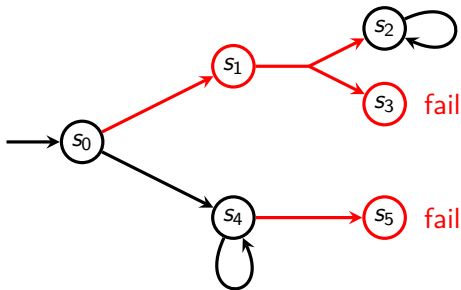


schmalz.cc/fond-fault-analysis

- Budde, C. E.; Dehnert, C.; Hahn, E. M.; Hartmanns, A.; Junges, S.; and Turrini, A. 2017. JANI: Quantitative Model and Tool Interaction. In *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, 151–168.
- Cascioli, L.; Jain, C.; Steinmetz, M.; Davis, J.; and Hoffmann, J. 2025. Safety Debugging of Tree-Ensemble Action Policies in AI Planning: From Fault Detection to Fault Fixing. In *Proc. of Workshop on Reliability In Planning and Learning (RIPL)*.
- Diekert, V.; and Kufleitner, M. 2022. Reachability Games and Parity Games. In *Int. Colloquium on Theoretical Aspects of Computing (ICTAC)*.
- Jain, C.; Sherbakov, D.; Vinzent, M.; Steinmetz, M.; Davis, J.; and Hoffmann, J. 2025. Policy Safety Testing in Non-Deterministic Planning: Fuzzing, Test Oracles, Fault Analysis. In *Proc. of European Conf. on Artificial Intelligence (ECAI)*.

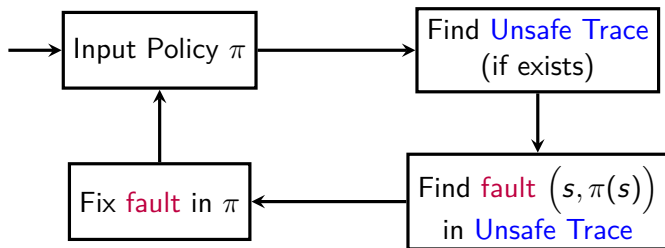
Propagate Unsafe Region based on [Diekert and Kufleitner, 2022]

Mark fail states as unsafe, then iteratively mark (s, a) as unsafe if some successor is unsafe, and mark s unsafe if all its actions unsafe



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- best and worst case $\Theta(|\mathcal{Y}|)$

$\mathcal{Y}$ are state-action pairs
with effect in unsafe
region

Context: Policy Repair Loop [Cascioli et al., 2025]

$(s, \pi(s))$ is a **fault** if s is safe but $\pi(s)$ leads to unsafe

Context: Problem Encoding

We use JANI and we may have exponentially-many initial states.